# Solution Requirements

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| **Date** | 30 June 2026 |
| **Team ID** | RW-2026-01 |
| **Project Name** | Rising Waters |
| **Maximum Marks** | 4 Marks |

**Define Functional and Non-Functional Requirements:**

**Functional Requirements (FR):**

Functional requirements define the core features and capabilities of the Rising Waters system. They describe how the system collects weather data, predicts flood risks, generates alerts, manages user access, and supports decision-making for effective flood management.

**Non-Functional Requirements (NFR):**

Non-functional requirements define the quality attributes of the Rising Waters system. They ensure the application delivers high performance, reliability, security, scalability, and usability while providing accurate flood predictions and uninterrupted service during critical situations.

Fill out the descriptions and necessary details for each category provided below.

**Step 1: Functional Requirements (FR)**

| SI.No | Requirement Category | Requirement Description | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Authentication | Users and administrators securely log in using username and password to access the system. | High |

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| **2** | Authorization Levels | *Role-based access control: Administrators can manage datasets, models, and alerts; Users can view flood predictions and warnings.* | High |
| **3** | External Interfaces | *Connects with weather APIs, rainfall sensors, river water level monitoring systems, and historical flood databases to collect real-time and historical data.* | High |
| **4** | Transactions Processing | *Collects weather data, preprocesses it, performs flood prediction using the machine learning model, and stores prediction results and alerts.* | High |
| **5** | Reporting | *Generates flood prediction reports, risk analysis dashboards, historical trend reports, and alert logs for administrators.* | Medium |
| **6** | Business Rules | *Flood alerts are generated when the predicted flood probability exceeds a predefined threshold. Alerts are categorized as Low, Medium, or High risk.* | High |
| **7** | Compliance to Laws or Regulations | *Complies with data privacy policies and government disaster management guidelines for handling environmental and public safety data.* | Medium |

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| **8** | Other | *Provides real-time flood notifications to citizens and emergency response teams through the alert system. (e.g., GDPR, HIPAA)]* | Medium |

**Step 2: Non-Functional Requirements (NFR):**

| **S.No** | **NFR Category** | **Requirement Description** | **Target Metric / Acceptance Criteria** |
|---|---|---|---|
| **1** | Performance & Speed | The system should process incoming weather data and generate flood predictions quickly. | Prediction generated within **2 seconds**. |
| **2** | Scalability | The system should support increasing volumes of weather data and multiple concurrent users. | Supports **10,000+ concurrent users** and continuous data streams. |
| **3** | Security & Data Privacy | Protects user credentials and stored data using secure authentication and encrypted communication. | Uses **HTTPS, AES-256 encryption, and secure authentication**. |
| **4** | Reliability & Availability | The system should remain operational during emergencies with minimal downtime. | **99.9% system uptime** with automatic recovery mechanisms. |

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| **5** | Usability & Accessibility | Provides an intuitive dashboard that is easy to use for administrators, disaster management authorities, and citizens. | Responsive UI compatible with desktop and mobile devices; follows accessibility best practices. |
| **6** | Other | The system should be maintainable, portable, and support future integration with additional weather sensors and machine learning models. | Modular architecture with easy deployment and maintenance. |